Entropy–Information Scaling of the Critical Coherence Index in Two-Dimensional ϕ^4 Field Dynamics

Dimensional Dependence and Robustness of Structural Instability

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Abstract

The Critical Coherence Index (CCI) has recently been proposed as a structural instability measure for nonlinear field systems, capturing the balance between dynamical activity and structural coherence [3]. In one-dimensional ϕ^4 field dynamics, the CCI was found to be consistent with an empirical scaling relation

$$\mathcal{C}_{\mathrm{CCI}} \sim \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}}, \qquad \alpha \approx 2.5,$$

where $\dot{S}_{\mathrm{cg}}^{(+)}$ is a coarse-grained histogram activity proxy and I_{nn} is nearest-neighbour mutual information [5].

In this work we test the robustness of this entropy–information scaling in two-dimensional systems. Using ensemble simulations of a two-dimensional ϕ^4 field, we find that the scaling persists in a strong monotonic sense, with Spearman correlation $r_s \approx 0.87$. Within the tested scan range, the optimal exponent shifts to $\alpha \approx 3.0$, suggesting a possible dimension-dependent modification of the scaling relation. A log-linear regression further shows that in two dimensions the CCI is more strongly controlled by the degradation of structural information than by histogram activity alone ($R^2 \approx 0.74$).

These results suggest that the CCI captures a robust mechanism of structural instability across nonlinear field systems, while its precise scaling reflects dimensional effects consistent with a dimension-dependent universality class. All results are empirical; a first-principles derivation of the scaling relation and its dimensional dependence remains an open problem.

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1 Introduction

Nonlinear field systems exhibit a wide range of dynamical behaviours, including coherent structures, metastable patterns, and fully chaotic regimes. While the underlying equations of motion define the space of possible configurations, they do not directly explain which structures persist and which decay.

In recent work [3, 4], the Critical Coherence Index (CCI) was introduced as a structural diagnostic on the solution manifold of field equations, quantifying the balance between destabilising dynamical activity and stabilising structural coherence (see Section 2 for the explicit definition). In one-dimensional ϕ^4 field theory, the CCI was found to correlate with an entropy–information ratio of the form $\dot{S}_{\mathrm{cg}}^{(+)}/I_{\mathrm{nn}}^{\alpha}$, with optimal exponent $\alpha \approx 2.5$ [5].

This result suggests that instability is governed by a competition between dynamical reorganisation and the preservation of structural information. However, it remained unclear whether this relation is specific to one-dimensional systems or reflects a more general principle.

The present work addresses this question by extending the analysis to two-dimensional ϕ^4 field dynamics. Higher-dimensional systems introduce additional degrees of freedom, richer pattern formation, and more complex pathways for energy redistribution, making them a natural testbed for robustness and possible universality.

The central question of this paper is:

> *Does the entropy–information scaling of the CCI persist in higher dimensions, or does it depend on dimensionality?*

We find that the scaling persists in a strong monotonic sense, but with a shifted effective exponent and a modified entropy–information balance, pointing toward a dimension-dependent universality class.

2 Model and Diagnostics

We extend the ϕ^4 scalar field model to two spatial dimensions:

$$\ddot{\phi} = \nabla^2\phi - V'(\phi), \qquad V(\phi) = \tfrac{1}{4}(\phi^2 - 1)^2, \tag{1}$$

on a periodic 48×48 lattice ($\Delta x = 1$), integrated with a fourth-order Runge–Kutta scheme ($\Delta t = 0.02$, $T_{\max} = 12$, seed $= 42$). Initial conditions are random mixtures of vacuum, domain-wall, localised, and noise components, as in the 2D companion study [2].

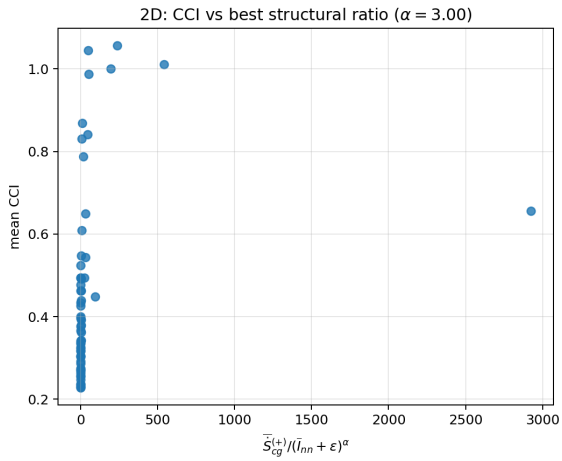
The Critical Coherence Index is [3]:

$$\mathcal{C}_{\mathrm{CCI}} = \frac{\Gamma_{\mathrm{inst}}\,\Gamma_{\mathrm{prod}}\,(1 + \kappa\, U_{\mathrm{struct}})}{\Gamma_{\mathrm{coh}} + \Gamma_{\mathrm{cons}} + \Gamma_{\mathrm{corr}} + \Gamma_{\mathrm{int}} + \varepsilon}, \tag{2}$$

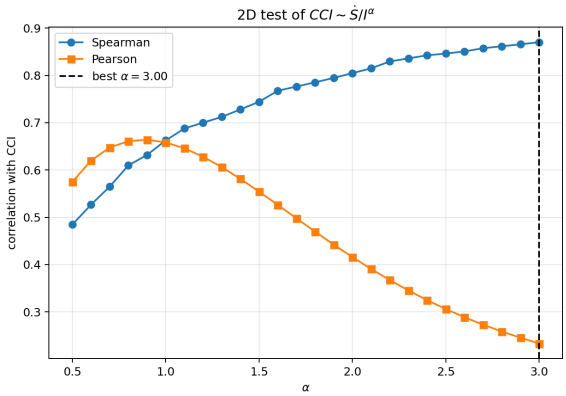
where $\Gamma_{\mathrm{inst}}, \Gamma_{\mathrm{prod}}$ are activity and instability diagnostics; $\Gamma_{\mathrm{coh}}, \Gamma_{\mathrm{corr}}, \Gamma_{\mathrm{int}}$ are coherence-preserving diagnostics; and U_{struct} measures sectoral imbalance.

The histogram activity proxy $\dot{S}_{\mathrm{cg}}^{(+)}$ and nearest-neighbour mutual information $I_{\mathrm{nn}} = I(\phi_i : \phi_{i+1})$ are computed on the 2D field configurations with spatially averaged diagnostics. The regime classification used in the phase diagram (Figure 1d) follows the empirical working thresholds of the programme, which should be understood as calibration benchmarks rather than universal constants.

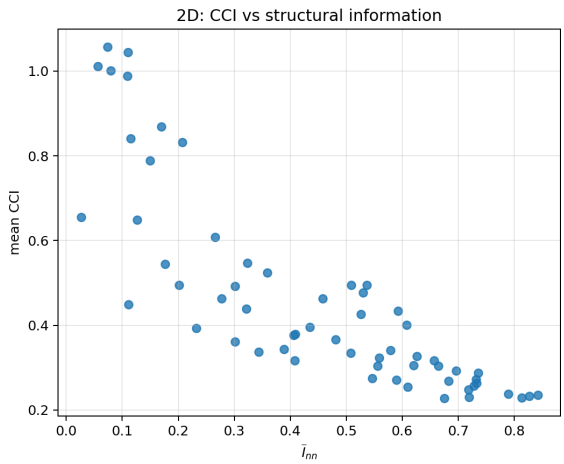
All code is available at [7].



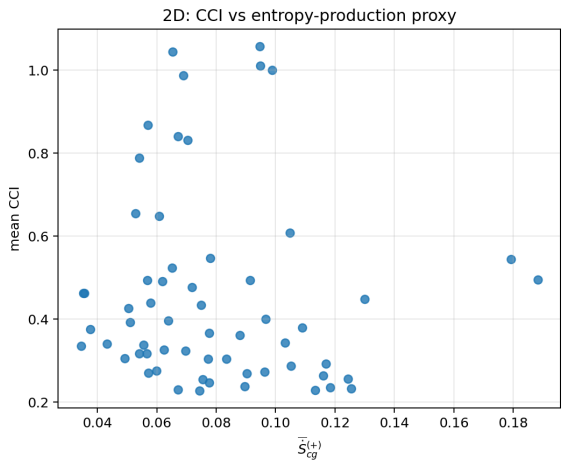
(a) Mean CCI.



(b) Mean structural free energy.



(c) Mean structural information.



(d) Dominant regime classification.

Figure 1: Phase-diagram view of structural instability in the 2D ϕ^4 ensemble as a function of initial momentum amplitude σ_π and noise amplitude σ_{noise}. The ordered, critical, and chaotic regions are consistently reflected across CCI, structural free energy, structural information, and dominant regime classification.

3 Results

Figure 1 shows the regime structure of the 2D ensemble in the control-parameter plane, while Figure 2 summarises the entropy–information scaling results.

3.1 Entropy–Information Scaling in Two Dimensions

We test the empirical scaling ansatz

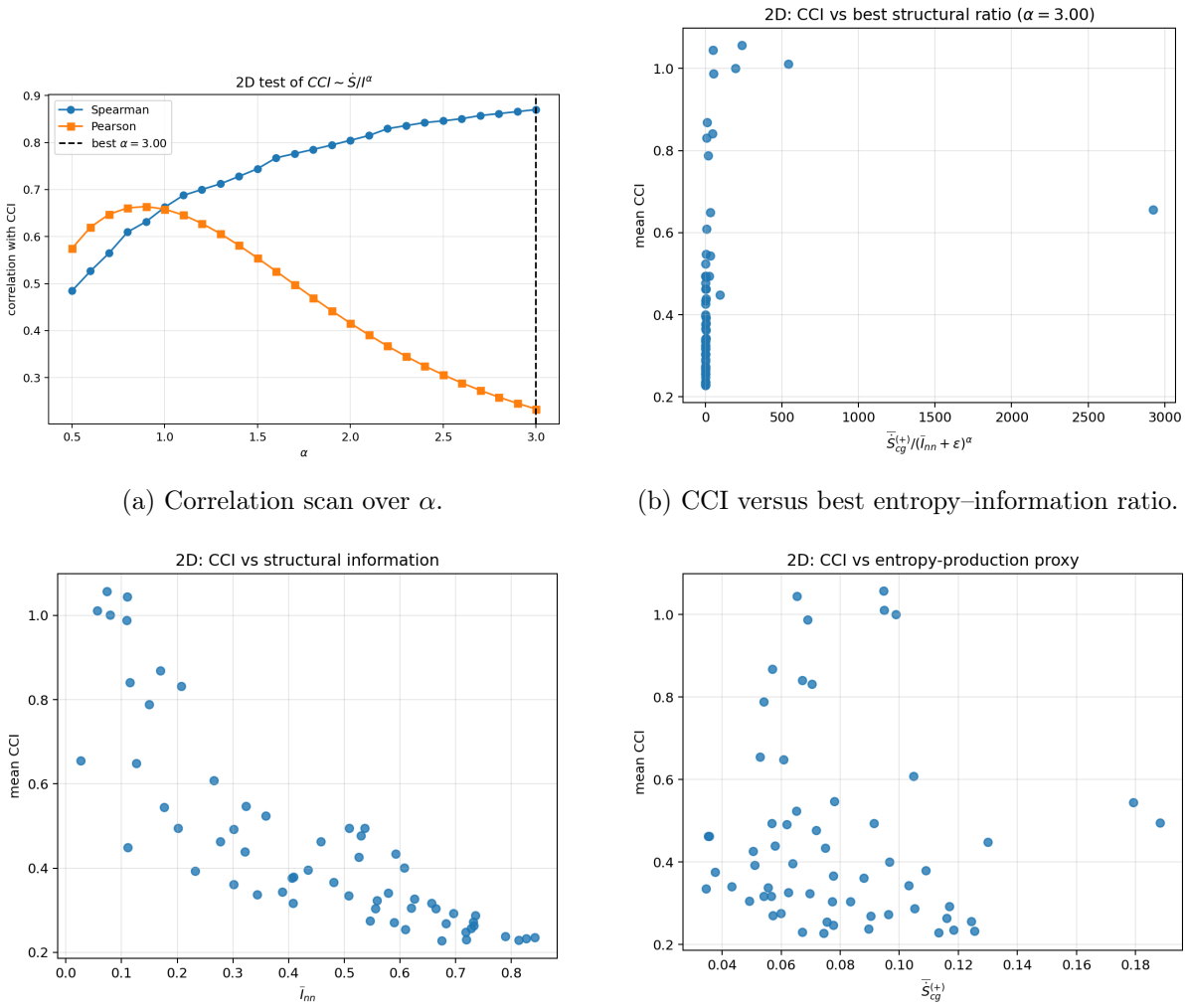
$$R_\alpha = \frac{\dot{S}_{\text{cg}}^{(+)}}{I_{\text{nn}}^{\alpha}} \tag{3}$$

by scanning $\alpha \in [0.5, 3.0]$ and computing the Spearman rank correlation between $\mathcal{C}_{\text{CCI}}$ and R_α.

Within the tested scan range, the maximum Spearman correlation is

$$r_s \approx 0.87, \tag{4}$$

attained at $\alpha \approx 3.0$. The corresponding Pearson correlation remains substantially lower ($r \approx 0.23$), confirming a strongly monotonic but distinctly nonlinear relationship.



(a) Correlation scan over α.

(b) CCI versus best entropy–information ratio.

(c) CCI versus I_{nn}.

(d) CCI versus $\dot{S}_{\mathrm{cg}}^{(+)}$.

Figure 2: Entropy–information scaling of the CCI in the 2D ϕ^4 ensemble. (a) Monotonic correlation increases across the tested range, reaching its maximum at $\alpha = 3.0$. (b) The best-fit ratio yields a strong monotonic but nonlinear relation. (c) CCI decreases with increasing I_{nn}, indicating coherence loss as the dominant instability mechanism. (d) Dependence on $\dot{S}_{\mathrm{cg}}^{(+)}$ alone is weaker.

3.2 Comparison with the One-Dimensional Result

The one-dimensional study [5] found $\alpha_{1D} \approx 2.5$. The two-dimensional results suggest:

$$\alpha_{2D} \approx 3.0 \quad \text{(within the tested range)}, \tag{5}$$

indicating that the entropy–information balance remains valid in higher dimensions, but the quantitative scaling is modified.

3.3 Role of Entropy and Structural Information

A log-linear regression of the form

$$\ln(\mathcal{C}_{\mathrm{CCI}}) = a_0 + a_1 \ln(\dot{S}_{\mathrm{cg}}^{(+)}) + a_2 \left(-\ln I_{\mathrm{nn}}\right) \tag{6}$$

yields $a_1 \approx -0.08$ (histogram activity, weak) and $a_2 \approx +0.49$ (structural information, dominant), with $R^2 \approx 0.74$. In two dimensions, the CCI is primarily controlled by the loss of structural

information; histogram activity is subleading.

3.4 Summary of Findings

- The empirical scaling of the CCI persists in 2D.
- The effective exponent shifts: $\alpha_{1D} \approx 2.5 \to \alpha_{2D} \approx 3.0$ (within tested range).
- Correlation: strongly monotonic ($r_s \approx 0.87$), nonlinear ($r \approx 0.23$).
- I_{nn} is dominant in 2D; $\dot{S}_{\mathrm{cg}}^{(+)}$ is subleading.

4 Discussion

4.1 Universality and Dimensional Dependence

The persistence of a strong monotonic relation between $\mathcal{C}_{\mathrm{CCI}}$ and $\dot{S}_{\mathrm{cg}}^{(+)}/I_{\mathrm{nn}}^{\alpha}$ in two dimensions supports the interpretation that the CCI captures a robust mechanism of structural instability. The shift $\alpha_{1D} \approx 2.5 \to \alpha_{2D} \approx 3.0$ indicates a dimension-dependent universality class: the qualitative structure is robust, but the effective exponent reflects dimensional effects.

4.2 Entropy vs Structural Information

The weak coefficient a_1 in the 2D regression suggests that histogram activity alone does not drive instability in two dimensions. Coherence loss appears to be the dominant mechanism, consistent with the picture that higher-dimensional systems provide more channels for structural reorganisation.

4.3 Relation to the Landau Framework

Within the Landau-type framework of [6], the order parameter $\mathcal{O} \sim I_{\mathrm{nn}}$ becomes increasingly relevant in higher dimensions. The shift in scaling exponent can be interpreted as a change in the effective dependence of the Landau coefficient on structural information.

4.4 Limitations

The precise value of α may depend on the entropy estimator, binning procedure, finite-size effects, and sampling resolution. The optimal α lies at the boundary of the tested range, indicating that a broader scan is required for precise determination. The CCI thresholds used for regime classification are empirical working values of the programme, not universal constants.

5 Conclusion

We have tested the entropy–information scaling of the CCI in two-dimensional ϕ^4 field dynamics. The empirical scaling relation $\mathcal{C}_{\mathrm{CCI}} \sim \dot{S}_{\mathrm{cg}}^{(+)}/I_{\mathrm{nn}}^{\alpha}$ persists with $r_s \approx 0.87$, supporting the robustness of the CCI as a structural instability measure. Within the tested range, the effective exponent shifts from $\alpha \approx 2.5$ (1D) to $\alpha \approx 3.0$ (2D), consistent with a dimension-dependent universality class. Structural information loss is the dominant contribution in 2D.

Future work will extend the analysis to 3D systems, alternative potentials, and reaction-diffusion models.

All results are empirical, derived from numerical simulations of a single field model. A first-principles derivation of the scaling relation and its dimensional dependence remains an open problem.

A Reproducibility

https://github.com/Chris4081/structural-selection-principle

Run:

```
pip install numpy pandas matplotlib scipy scikit-learn
python cci_entropy_scaling_2d.py
```

Key parameters: $N = 48 \times 48$, $\Delta t = 0.02$, $T_{\max} = 12$, seed $= 42$, $\alpha \in [0.5, 3.0]$.

References

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